Word-count declaration

Manuscript: *A Comparability Protocol for Benchmarking Relational Database Access Layers in Express.js* (submission build `ist/ist_main.tex`).

Counted with `texcount` on the seven body sections plus the structured abstract. The IST *Guide for Authors* sets the limit for a research paper at **15,000 words**, and states that **"references and appendices are part of the submission and count against the total number of words, and figures and tables count 200 words each."** This count applies that rule exactly: the reference list is included and each of the eight main-text floats counts 200 words. The paper has **no appendices** — all supplementary material is in the separate `supplement.pdf`, which under Elsevier's policy is online supplementary material, not an appendix, and is not part of the manuscript word count. The mandatory declaration sections (CRediT, Declaration of competing interest, Funding, Data availability, and the generative-AI declaration, ~300 words) are required metadata, not article content or appendices, and are excluded per Elsevier convention.

Component	Count
Body text (7 sections)	10649
Structured abstract	299
Tables and figures in the main text (8 × 200)	1,600
Reference list (62 entries)	1,848
Total (IST rule)	**14396**

This is under the journal's 15,000-word limit (with the abstract counted; excluding the abstract it is 14097). The structured abstract is 299 words (under the journal's 300-word structured-abstract limit). Thirty-two tables and two figures are placed in the numbered online supplement (`supplement.pdf`, Supplement Tables S1–S32 and Supplement Figures S1–S2), which is submitted with the manuscript and archived under the same Zenodo DOI as the code and data; supplement floats are not part of the main-text count. A later revision added per-cell bootstrap 95% CIs to the p99 column of every pattern table and a paired p99 significance table (Supplement Table S30), delivering the p99 inference the outcomes table lists as primary; these widen existing tables and add one supplement float, not a main-text float.

In this revision (round 5) the entire benchmark was re-run on the latest stable library versions (Prisma 7, Express 5, etc.); the manuscript was repositioned as a reproducible benchmark and a configuration-specific comparison; the body prose was cut by roughly 1,800 words to remove repeated conclusions and filler (the five recurring findings are now stated once, in the results section, and cross-referenced elsewhere); the secondary access patterns (point read, keyset range scan) were reduced to lean summaries with their full tables in the supplement; the controlled/varied/uncontrolled factor table was moved to the supplement (Table S27); and five peripheral citations (general NoSQL and OLTP benchmark context) were dropped, taking the reference list from 70 to 62 entries. Six single-host experiments added earlier in the round (utilization-controlled open-loop, multi-worker cluster, per-layer pool frontier on both engines, mixed read/write, replicated fan-out, and a longer-run sensitivity check) remain as Supplement Tables S21–S26 with a one-sentence pointer each in the body.

Two later patch revisions left the reference list at 62 entries: the re-review revision (v1.5.1) dropped one peripheral citation, and v1.5.2 added one methodological citation (Papadopoulos et al. 2021, *IEEE Transactions on Software Engineering*, on reproducible performance-evaluation principles) anchoring the paper's perishability thesis. That citation adds no body prose (a bare `\{}cite`), so the only change to the count is the single reference entry (reference list 1,842 → 1,848 words, total 14,917 → 14,923).

Revision round 6 retracted the causal same-SQL attribution (now a bounding contrast), renamed the "idiomatic" estimand, separated the capacity/overload/matched-utilization quantities, tempered the perishability claim, and added the required statistical disclosures (Kendall's τ, an ex-MikroORM robustness note, a layer$\times$engine interaction table, and an insert-dispersion figure, all in the supplement). The condensed Prisma case study and the removal of repeated restatements offset most of these additions, so the body changed marginally (11,376 to 11,388) and the abstract was tightened from 313 to 298 words; the total is 14,934. The longer-window tail sensitivity (Table S23), interaction magnitude (Table S28), and insert dispersion (Figure S1) are the new supplement floats.

A subsequent round-6 clarification renamed the primary 50-connection matrix from "saturated" to a "fixed 50-connection high-load operating point," since the throughput knee is demonstrated (Supplement Figure S2) only for the deep fetch, while the ten-connection pool—not a per-pattern knee—is the binding high-load constraint (Section 3, Controls); the per-layer/pool/open-loop "saturat*" uses are unchanged. The rename plus its one-clause justification added a net 15 body words (offset by tightening three tail-latency sentences) and one abstract word, taking the body to 11,444 and the total to 14,991.

A further round-6 clarification softened the causal claim that each layer is "single-core-bound" / "multi-core-bound": the equal-CPU and multi-worker controls show that extra application cores do not raise a process's throughput (so the ranking is not a core-budget artifact), but do not prove single-core CPU is the causal bottleneck. The label was replaced by that observation at every prose site and in the equal-CPU and cluster table captions (the latter also corrected: its own data shows Prisma flat across 1/2/4 cores, so "Prisma requires roughly four cores to approach pg" was wrong—the four are cluster workers, not cores). Merging a now-redundant sentence offset the reframe, so the body is 11,443 and the total 14,990; the CPU-utilization measurements ("$\{}approx$110\{}% of one core") are unchanged.

A final round-6 clarification separated the *fact* that an earlier Prisma result did not reproduce on the re-run from the *hypothesis* that Prisma's version/architecture change caused it: the Discussion now states the non-reproduction as a fact and its cause as an explicitly hedged hypothesis (the vanished premium was Prisma-specific and only Prisma changed architecture, so its Rust-free rewrite is the likeliest driver, but the re-freeze advanced the whole toolchain at once, so it cannot be isolated), and the intro/conclusion headline sentences now say the result was retired "on newer versions" rather than attributing it to "Prisma's Rust-free successor." Trimming the paragraph's lead-in and a parenthetical offset most of the hedge, taking the body to 11,447 and the total to 14,994.

A final round-6 terminology change replaced the over-claiming label "idiomatic" (~53 occurrences) with "documentation-selected" throughout, with lighter synonyms ("documented default", "default") where it read clunkily. The methodology already disclaimed the connotation ("Documentation order is a reproducible selection heuristic, not evidence that the chosen API is performance-optimal or the most common production choice"); this makes the terminology consistent with that definition. The "default-configuration" estimand name is unchanged; the generated table captions (sameplan, altloading, insert-dispersion) and the replication-package `METHODOLOGY.md` were updated to match. Dropping the now-redundant "used idiomatically" disclaimer clause took the body to 11,437 and the total to 14,984.

A final round-6 change simplified the statistical emphasis: the results subsection "Measurement stability and significance" was renamed "...and effect sizes" and rewritten to lead with geometric-mean ratios, bootstrap CIs, and practical significance (the narrowest adjacent step, 1.05, sits at the $\pm$5% equivalence margin — statistically distinguishable yet practically marginal; the up-to-7$\times$ spread is decisive), with the post-hoc adjacent-pair permutation/Wilcoxon/Bonferroni tests compressed to a secondary, descriptive robustness note. The conclusion, methodology, and threats sentences were aligned to the same hierarchy (effect sizes and intervals primary; adjacent-rank tests secondary and descriptive). No statistic was added or removed; the significance table (Table 7) stays in the main text. Compressing the p-value enumeration offset the added practical-significance prose, so the body is 11,435 and the total 14,982.

A follow-on round-6 change reported the layer×engine interaction by its magnitude rather than its p-value: the RQ2 paragraph now leads with the per-layer PostgreSQL÷MySQL throughput ratios (reads a narrow $\{}approx$1.0–1.6× band, the insert scattering 1.6–3.2× and reordering layers across engines), with the blocked permutation test demoted to confirming the interaction is nonzero (its floor p says nothing about size); the outcomes-table row foregrounds the effect ratios (Table S28). The added magnitude detail took the body to 11,444 and the total to 14,991.

A final round-6 change softened the novelty claims to match the (explicitly non-systematic) scoping search: the one bald claim, "the combination missing from prior art" (introduction), became "a combination not identified in our documented search," and "a design choice absent from every access-layer benchmark surveyed above" (related work) became "not found in the access-layer benchmarks surveyed above." The four-axis gap statement and the related-work positioning were already scoped ("In the sources we searched. . . "; "a structured scoping search (not a systematic review)"). Net +3 words; the body is 11,447 and the total 14,994.

A final round-6 change stated explicit library inclusion/exclusion criteria in the methodology (each layer is maintained, in common use, occupies a distinct taxonomy tier, and — for the portable tiers — runs against both engines), with the reasons the named candidates were omitted: PostgreSQL-only clients (Slonik, pg-promise) are non-portable, and Kysely is a query builder already represented by Knex. The full criteria and a per-candidate table are in the replication package (`experiments/METHODOLOGY.md`, not word-counted). The addition was offset by removing a redundant tier-classification clause (Drizzle's placement, already given in the layer list) and tightening the vendor-independence sentence, so the net change is +1 word: the body is 11,448 and the total 14,995.

A final round-6 change added a short **artifact-reproducibility table** (Supplement Table S31: version/commit and DOI, software and hardware requirements, the run commands, the time and resources, and which results are regenerated automatically from the archived raw data), with a one-line pointer in the Data Availability section. Both are outside the counted body — the table is a supplement float (uncounted) and Data Availability is back-matter, not one of the seven counted sections — so the declared total is unchanged at 14,995. That round-6 change brought the supplement to thirty-one tables (S1–S31); the table was placed last, so S1–S30 kept their numbers.

A final round-6 pass shortened the manuscript and fixed editorial errors. Repeated conclusions were consolidated to one canonical statement each with cross-references — the Prisma retired-headline story, the version-sensitivity thesis, and the same-SQL "bounds but does not isolate" caveat were each stated once and cross-referenced elsewhere — and the most duplicative auxiliary prose (the open-loop/utilization tail block and the thrice-told multi-worker cluster result) was compressed to its conclusion plus the Supplement table pointer; every trimmed number still appears once in its Supplement table. This cut the body by 232 words, from 11,448 to 11,216, and the total from 14,995 to **14,763** (237 words of headroom under the 15,000 limit). Editorially, the run-in `\{}paragraph` headings were rendering a double period under `elsarticle` (which auto-appends a period to a heading that already ended in one — "the instrument is the contribution.."); the trailing period was removed from all twelve headings so each now renders a single period.

A round-7 methodology-and-scope pass sharpened the claims without changing any measurement: the estimand was renamed from "default-configuration" to "documentation-selected implementation-and-strategy" (choosing the first documented API is not the library's default configuration); the three claim levels — the primary implementation-and-strategy comparison, the same-SQL raw-path bounding control, and the individually non-identified mechanisms — were named and kept separate; "capacity", "saturating throughput", and "overload" were reserved for the deep-fetch sweep, where a throughput knee was actually measured, and the primary matrix's fixed point is described as "high-load" rather than "overload"; and a compact **experiment-scope table** (Supplement Table S32: each experiment's access pattern(s), engine(s), layer count, and repeated runs) was added last, so S1–S31 keep their numbers. The supplement now holds thirty-two tables (S1–S32). A companion CPU-and-statistics pass in the same round demoted the equal-CPU experiment to an exploratory sensitivity check and corrected its "at most a few percent" wording (the ORM per-core values wobble 10–17% run-to-run); added that the bootstrap intervals capture within-campaign variability, not cross-machine generalization; noted that 1–2 ms p99 gaps sit at the millisecond measurement floor; and brought the raw MySQL-insert distributions (Supplement Figure S1) into the body. A final conclusions-and-language pass then limited "access-layer overhead" wording to the combined implementation-and-strategy quantity the design actually measures, replaced the "thin layers are a safer default" recommendation with "benchmark the relation-heavy hot path for the specific application", and added an explicit "configuration-specific" label to the Conclusion (the cautious "in the sources searched" novelty phrasing was already in place). A closing presentation-and-control pass then removed repeated caveats — the standalone same-SQL "bounds not isolates" Discussion paragraph (already stated in Results, the abstract, and the Conclusion), the duplicated horizontal-scaling and "productivity/type-safety not measured" sentences in Practical Guidance, and a redundant single-host restatement — and, in the abstract, dropped the secondary Kendall-tau and matched-utilization results while scoping the open-loop and equal-compute-budget conditions to the deep fetch. A pre-submission consistency sweep confirmed the pinned library versions match `package.json`, the DOI and GitHub link resolve, and every headline number matches its table. That sweep also corrected a CV figure ("within 4.2%", not 4.0%, on the reads), removed the redundant per-cell median CIs from the main-paper paired significance table (Table 7) that had disagreed with the pattern table's bounds, and reconciled two loose prose phrasings ("the faster layers" cluster within 5%; the transactional ordering "broadly tracks" the single-row insert apart from Prisma). These are all rewordings, so the manuscript total was **14,903** words (11,356 body, 97 under the 15,000 limit); the structured abstract is 292 words.

A subsequent editorial pass condensed the main text by roughly **24%** (body 11,356 → 8,669 words; main paper 51 → 43 pages) so it reads as "a reproducible benchmark methodology demonstrated through a dual-engine case study," with the full audit trail in the supplement. Nothing was deleted: the detailed statistical construction (permutation/bootstrap/TOST/ANOVA) became the supplement's *Statistical methods* section, the measurement internals (warm-up, order-invariance, tail estimation, resource sampling) its *Measurement details* section, and the four-pitfall checklist its *Benchmarking-pitfalls checklist* section (main text keeps a named summary); Methodology, Results, Discussion, and Threats were tightened and repeated single-host/version-sensitivity caveats consolidated. The manuscript total is now **12,216** words (8,669 body + 299 abstract + 1,848 references + seven main floats), well under the 15,000 limit; the supplement grew from 26 to 29 pages. All citations were kept in the main text (the supplement has no bibliography), so the reference list is unchanged. This revision is archived as release v1.6.4 (DOI 10.5281/zenodo.21444624).

A round-8 revision subordinated the whole narrative to one primary contribution — a **comparability protocol** for access-layer benchmarking (a semantic-equivalence gate, a documentation-selected estimand bounded by a same-SQL control, and separated operating points) — explicitly distinguished from the reusable artifact and the empirical case study (title, abstract, introduction contribution list, related-work positioning, discussion, conclusion, and highlights). It also addressed the estimand's construct validity (the documentation-selected treatment is a reproducible persona proxy, not a claim about practitioner choice; a same-SQL control that *bounds* rather than *decomposes* the strategy contribution; new evidence in Supplement Table S33), rewrote the same-SQL result in consistently non-causal terms and diagrammed its changed components in a new main-text table (Table 6), and reweighted the RQ2 cross-engine treatment to lead with the concrete rank reversals and the layer×engine interaction magnitudes (on the insert \{}texttt{knex} 1→3 and \{}texttt{drizzle} 2→1 trade the lead while Prisma falls 4→last; the per-layer PostgreSQL÷MySQL advantage scatters 1.6–3.2× on the insert and 1.4–1.9× on aggregation; Supplement Tables S27–S28) rather than characterizing the n=7 Spearman coefficients with qualitative labels such as "moderate," which were removed; the coefficients are now reported only as coarse descriptive summaries. These changes added a main-text component diagram (seven → eight main floats) and expanded the estimand/RQ2 prose; the body is 9,648 words and the total 13,395, and the structured abstract was tightened (a duplicated version-sensitivity sentence dropped, the reversal detail added) to stay under 300.

A round-8 workload-scope pass then narrowed the one generalizing statement the reviewer flagged — "the ORM penalty concentrates on relation-materializing traffic" — to the tested implementations and patterns: the Discussion now says the documentation-selected ORM spread was largest on the single deep/nested-fetch shape tested and adds that one nested-fetch topology (the fan-out sweep varies its breadth, 0–500 children, but not its depth, schema, or query mix) cannot establish a general distribution of ORM penalties, so the pattern is a benchmark observation to be re-measured on the target workload, not an established law; the Conclusion and the practice paragraph were scoped to "the five tested patterns" / "these probes" to match. The added scoping took the body to 9,739 words and the total to 13,486, still under the 15,000 limit.

A round-8 horizontal-scaling pass then weakened the claim, flagged by the reviewer, that a layer's low per-process throughput is "recoverable by horizontal scaling." The node-cluster check (Supplement Table S24) scales only 1→4 workers on a co-located 16-core host, measures no database-side saturation, and on MySQL the native driver keeps scaling so the gap widens; the one PostgreSQL knee (native `pg`) was previously attributed to becoming "database-bound" with no DB-side evidence. Every claim site (Results, Discussion ×2, Threats, the Multi-worker-scaling supplement section, and the S24/S4 table captions in their generators and both copies) now reads "may be addressed by additional application processes until another bottleneck is reached," scoped to the tested 1→4-worker range, with the unsupported "database-bound" attribution removed and the MySQL counter-case and the absence of DB-side saturation measurement stated explicitly. The added caveats took the body to 9,879 words and the total to 13,626, still under the 15,000 limit.

A round-8 language pass then matched the strength of the prose to the replication of the low-run secondary experiments the reviewer flagged (equal-CPU three runs, pool-size three runs, transactional write five runs, and by the same principle the node-cluster and open-loop coordinated-omission checks). Inferential verbs became consistency statements: "an equal-CPU check confirms no layer converts extra cores" -> "is consistent with no layer converting"; the pool-size frontier "shows the ordering preserved" -> "is consistent with the ordering being preserved" and "does not drive the ranking" -> "does not appear to drive the ranking" (in the Threats prose, both supplement pool-size sections, and the S4/S7/S25 captions in their generators and copies); the equal-CPU caption's "is not an artifact of unequal core budgets" -> "does not appear to be an artifact"; the open-loop CO cross-run "confirms the tail ranking holds" -> "is consistent with the tail ranking holding"; the cluster and open-loop "shows" -> "indicates"; and the transactional-write section now states its "median of five runs, exploratory" replication; and the alternative-eager-loading check (median of five) "rules out a mere default-strategy artifact" -> "argues against" and its "deficit is not an artifact of a poor default" -> "does not appear to be an artifact" (Results, Threats, the construct-validity supplement section, and the

S18 caption in its generator and both copies). Deterministic checks (byte-identity, checksums) and the primary-rigor claims (the 25-run matrix, its permutation test, the 25+10-run durability secondary, and the multi-point concurrency sweep's "not an artifact of a single point") keep their stronger wording. The rewording took the body to 9,891 words and the total to 13,638, still under the 15,000 limit.

A round-8 minor-concerns pass addressed five readability/labeling points. The abstract was lightened (MC1/MC4): its Methods clause dropped "equal-compute-budget conditions" (the equal-CPU check is an exploratory secondary), and its Conclusion dropped the retired-five-core-Prisma parenthetical (that result is stated in full in the Discussion and covered by the temporal external-validity threat), so the abstract stays comfortably under the 300-word limit and the core deep-fetch result reads more prominently. "Vendor-independent"/"vendor independence" applied to the study and the coverage axis became "independently authored"/"independent authorship" (MC2), since the manuscript means authorship, not design, independence (the "No (vendor)" prior-art column is unchanged). Drizzle's category label "orm-lightweight" was merged into "orm" (MC3) — an interpretive performance-implying label on an experimentally untested taxonomy — in the Category column of the six pattern/resource tables (both copies), the two generators (`ci-tables.mjs` CAT map and `analyze.mjs` display normalization) and `config.mjs`, with the methodology prose softened to "a query-builder-style ORM"; raw.json's internal category field is left as-is (not reader-facing, so no checksum churn). No secondary material was moved to the supplement (MC5): the manuscript is under the word limit, so density, not length, was the concern, and the R8 caveats are load-bearing for the other points; the abstract trim is the density fix. The body is 9,890 words and the total 13,637, still under the 15,000 limit.

A round-8 novelty-discipline pass held the strongest novelty claim to what a scoping (not systematic) search supports and distinguished the two kinds of novelty. In the Positioning, "none combines [the four properties]" became "we did not identify a study combining [them] — each we found covers at most two or three," followed by an explicit statement that the paper claims only "we did not identify a prior study combining these properties in the documented search," not priority the scoping search cannot establish. In the Introduction's contribution paragraph, the case study is now framed as contributing novel experimental evidence (a dual-engine, throughput-and-p99, documentation-selected comparison not identified in the searched sources) that is configuration-specific, distinct from the novel benchmark infrastructure (the protocol and its harness), which is the more durable contribution. No "first"/priority claim is made anywhere; the audit found only enumerative/temporal "first"s. The introduction's four-axis gap statement was also tidied so the "four coverage axes" are named explicitly (access-layer taxonomy, engine, joint throughput/tail metrics, independent authorship) and the separate HTTP-framework/client-observed gap is presented as "beyond these four axes" rather than as a fourth axis, matching the prior-art table's four columns. The additions took the body to 9,982 words and the total to 13,729, still under the 15,000 limit.

A round-8 closing pass responded to the reviewer's summary and the point-12 list. All six Essential items were already addressed in R8 points 1–7 (sharpen contribution; narrow the five-pattern-workload claims; weaken horizontal scaling; same-SQL as a compound intervention; reduce the weight on the n=7 rank correlations; downgrade low-replicate sensitivity conclusions), and the strongly-recommended distributional write presentation (Supplement Figure S1) and artifact-provenance table (Supplement Table S31) are in place. "Documentation- selected" is kept: the construct-validity treatment already disclaims representativeness, so the term is justified, and a second rename would risk reopening the estimand discussion. For the "over-engineered rhetoric" concern the main text got a light, safe streamlining that removed no load-bearing caveat: the duplicated "productivity/type-safety not measured" qualification in the Practical-guidance paragraph is now stated once, and a few redundant parenthetical "(Section~. . .)" cross-references in the densest sections (Results, Threats) were dropped where the same section was already cited nearby. No section was reordered and no float moved (the deeper-restructuring option was declined), since the caveats are load-bearing for the Essential points. The trimming took the body to 9,965 words and the total to 13,712, still under the 15,000 limit.

The round-8 revision (all of the above from point 1 through the light streamlining) is archived as release v1.6.5 (DOI 10.5281/zenodo.21455542).

A round-8 follow-up (reviewer point 6.1) stopped calling the same-SQL result a "bound." The reviewer noted that the default-path and raw-path measurements evaluate two different combinations of factors, so — absent interaction assumptions — no ordering theorem makes the raw-path spread a bound on the intrinsic library effect (raw mode may interact differently with each library and may bypass the very mechanisms one wants to characterize). Following the reviewer's strong recommendation, every same-SQL "bound/bounds/bounded" was reworded to a **standardized (same-SQL) contrast / diagnostic** across the abstract, introduction, methodology, results (including the Table 6 caption and the `tab:sameplan` generator + both copies), threats, discussion, conclusion, related work, and supplement; a one-sentence caveat was added in Results ("Because raw mode changes several mechanisms at once and may interact differently with each library, this residual is a diagnostic contrast, not a bound on the intrinsic library effect — no ordering theorem holds without assumptions about interaction effects"). The compound-intervention framing (it changes several mechanisms together; Table 6) is unchanged; only the inequality/ordering implication is removed. The added caveat took the body to 10,084 words and the total to 13,831, still under the 15,000 limit; the structured abstract is 297 words.

The point-6.1 follow-up (same-SQL standardized contrast, not a bound) is archived as release v1.6.6 (DOI 10.5281/zenodo.21456366).

A round-8 follow-up (reviewer point 6.2) strengthened the semantic-equivalence gate rather than renaming it. The reviewer noted that finite-probe byte-equality is not full implementation equivalence and that a non-2xx check catches an HTTP 500 (the MikroORM case) but not a silently incorrect write. A new harness check, `bench/verify-writes.mjs`, validates post-write database state through an independent native-driver connection, for every adapter on both engines: the single-row insert wrote exactly the requested field values and generated identifier (one `posts` row, no stray comments); the transactional write inserted the post and its five comments with the exact fields (one `posts` and five `comments` rows — one intended logical operation); and a transaction whose last comment violates the author foreign key throws and rolls back with no partial state. All adapters pass on both engines, so no throughput number changes. Methodology now describes the write gate and adds an honest finite-probe caveat for reads; the Introduction contribution bullet and the Discussion MikroORM note were updated; `REPRODUCE.md` and the artifact-reproducibility table (Table S31) add the `verify-writes.mjs` command. The added description took the body to 10,264 words and the total to 14,011, still under the 15,000 limit.

The point-6.2 gate strengthening is archived as release v1.6.7 (DOI 10.5281/zenodo.21457569).

A round-8 follow-up (reviewer point 6.3) specified the comparability protocol normatively, abstracted from the case study. A new Methodology subsection, "The comparability protocol" (`sec:protocol`), states it independently of the Express/access-layer setting in five parts: **Inputs** (treatments, workload, output semantics, operating-point definition); **mandatory stages, in order** (correctness oracle → treatment-definition rule → strategy-control design → capacity identification → demand/utilization experiments); **pass/fail** cell-admission (a cell is admitted only if the oracle passes — equivalent non-mutating outputs and a correct post-write state; non-equivalent output, an invalid write, or a degenerate plan disqualifies it); **outputs and their interpretation** (descriptive of the treatment-and-strategy, not a causal decomposition; rankings configuration- and version-specific; only relative within-condition differences travel); and **applicability limits** (byte-equality is a valid oracle only for deterministic, canonically serializable outputs — for unordered collections, floating point, timestamps, nondeterministic identifiers, or differently serialized equivalents the oracle must be a semantic comparator). The Introduction now points to this specification. The added subsection took the body to 10,649 words and the total to 14,396, still under the 15,000 limit.

The point-6.3 protocol specification is archived as release v1.6.8 (DOI 10.5281/zenodo.21457745).

A round-8 follow-up (reviewer point 6.4) narrowed the RQ1 construct and added a performance-conscious co-primary regime. **Part A** reworded the GQM goal from "quantifying its performance cost" to "quantifying the performance of its documentation-selected implementation strategy," tying RQ1 to the selected strategy (the driver-versus-ORM asymmetry — native hand-SQL versus the ORM's documented relation loader — was already stated in Study Design). **Part B** added, on the deep fetch (the only one of the five patterns exposing a documented strategy choice), a second co-primary regime: the performance-conscious regime takes the faster of a layer's two documented loading strategies, admitted only where byte-identical to the documentation-primary output under the `bench/verify.mjs` oracle (a defined, narrow tuning budget that never rewrites SQL, adds caching, or changes the schema). A new harness (`scripts/deepfetch-regimes.mjs`, a warmup pass then 25 timed repeats each) measures both regimes for the four data-mapper ORMs on one footing (`results/deepfetch-regimes.json`, added to the checksum manifest — now thirty-four raw-data files); Study Design defines the tuning budget, Results reports the two regimes, and a new supplement float (Table S34) tabulates them. The finding reinforces rather than overturns the documentation-primary reading: for Sequelize and MikroORM the documented join is already the faster strategy (the two regimes coincide), only Objection on MySQL gains (its documented single-join alternative beats its select-in default), and even there the performance-conscious ORM deep fetch stays below the native driver (a same-harness native reference row, 3,347/2,274 req/s, is included in Table S34). The added prose took the body to 11,008 words and the total to 14,755, still under the 15,000 limit; the structured abstract is unchanged at 297 words. The supplement now holds thirty-four tables (S1–S34) plus two figures.

The point-6.4 follow-up (narrowed RQ1 construct + performance-conscious co-primary regime) is archived as release v1.6.9 (DOI 10.5281/zenodo.21459069).

A follow-up (reviewer point 6.5) demonstrated **capacity identification across all five patterns**, not only the deep fetch. The operating-point protocol requires that capacity be identified so the fixed 50-connection point can be read against it, but capacity sweeps previously existed only for the deep fetch, leaving the relative utilization of the other four patterns unmeasured. A per-pattern concurrency sweep (`bench/scaling-patterns.mjs`, connection ladder 1–200, every layer, both engines → `results/scaling_patterns.json`, the thirty-fifth raw-data file) locates each pattern's throughput knee at or below 50 connections, with median utilization at the 50-connection point of 97–100\{}% across layers (minimum 82\{}%); the fixed point therefore places all five patterns at high utilization of their own capacity, so the cross-pattern p99/spread comparison is one at comparable, measured relative utilization rather than unknown fractions of capacity. Study Design and the RQ3 results now state this, the new Supplement Table S35 reports the per-pattern knee and utilization, and the equal-utilization open-loop tail and exploratory equal-compute check are labeled as remaining demonstrated on the deep fetch. The added prose took the body to 11,156 words and the total to 14,903, still under the 15,000 limit; the structured abstract is unchanged at 297 words. The supplement now holds thirty-five tables (S1–S35) plus two figures.

The point-6.5 follow-up (per-pattern capacity/utilization sweep) is archived as release v1.7.0 (DOI 10.5281/zenodo.21461236).

Highlights (5 bullets, each ≤ 85 characters) are in `highlights.tex`.